

Sepehr Mahmoudian

SENIOR AI ENGINEER

Machine Learning • Knowledge Graphs • Multimodal & Embodied AI

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Berlin-Brandenburg • available from Oct 2026 • German citizen • EN fluent / DE good / Persian native



PROFILE

Senior AI Engineer with 10+ years across research and production. Builds Python/Rust systems for knowledge graphs and Graph RAG, multimodal AI and embodied AI, LLM/VLM pipelines, Vision-Language-Action (VLA) evaluation and computer vision.

LLMs / VLMs • Knowledge Graphs • VLA Evaluation • Model Evaluation

OPEN SOURCE • DATA PROVENANCE • MODEL VALIDATION • AI SYSTEMS

AI ENGINEERING

10+ YEARS

RESEARCH TO PRODUCTION

PYTHON • RUST • AI / ML

PROFESSIONAL EXPERIENCE

Independent AI/ML Engineer & PhD Candidate

2020.04 - Present

Self-employed - PhD candidate at Technische Universität Darmstadt

• Berlin & Darmstadt, Germany

- Building Engram, Prisoma, CREBAIN and Manwe: Python/Rust research systems for evidence-linked neural models, auditable VLA diagnosis, multimodal 3D perception and model-export validation
- Publishing open-source models and datasets on GitHub and Hugging Face, including the DOI-backed Cobot Atlas; completing a PhD dissertation on Partial Information Decomposition (PID)

Senior AI Engineer

2026.01 - 2026.03

Neural Frames UG

• Berlin & Hamburg, Germany

- Evaluated LLMs, VLMs and end-to-end pipelines, tracing runs and results with Langfuse
- Contributed to Python pipelines for model evaluation and inference

AI Engineering Resident (Hacker Room)

2025.09 - 2026.01

Merantix AI Campus

• Berlin, Germany

- Built Dreamweave, a generative-media system combining Gaussian splatting and diffusion models for spatial and temporal anchoring
- Implemented confidential-computing smart contracts in Rust under a NEAR Foundation contract
- Deployed product services and data infrastructure across Cloudflare Workers, D1, R2, Google Cloud Run and AWS EC2

Senior AI Engineer

2024.07 - 2025.07

Vision Labs UG (haftungsbeschränkt)

• Berlin & Hamburg, Germany

- Built and fine-tuned multimodal language-and-vision agents for client applications in Python
- Established prompt optimization, LLM evaluation and observability workflows with DSPy and Langfuse
- Designed data pipelines and MLOps across Azure and Google Cloud with staged environments, automated tests and CI/CD
- Led feasibility assessment, architecture and delivery for client AI projects

Research Associate

2021.07 - 2024.07

Freie Universität Berlin

• Berlin, Germany

- Scaled and optimized large neural-network simulations in C/C++ and Python
- Built retrieval-augmented generation (RAG) workflows for technical literature and evaluated AI-agent reliability
- Applied statistical and machine-learning analysis in Python; reviewed code and research manuscripts in an international team

Research Associate

Georg-August-Universität Göttingen • Göttingen, Germany

Neural networks and reproducible data workflows

2019.07 - 2020.06

Research Associate

Goethe-Universität Frankfurt • Frankfurt am Main, Germany • Simulation and statistical analysis of neural-network models

2017.03 - 2019.07

Visiting Scientist

Max Planck Institute for Dynamics and Self-Organization • Göttingen, Germany • Statistical methods for neural-network analysis

2017.03 - 2019.03

Research Associate

Forschungszentrum Jülich • Jülich, Germany • Neural-network simulations and reinforcement-learning experiments

2015.02 - 2017.03

Software Developer

Centre for Content Creation • Cyberjaya, Malaysia • Award-winning iPhone/iPad applications and digital products

2010.09 - 2012.03

EDUCATION

PhD Candidate (Dr. rer. nat.)

Technische Universität Darmstadt • Darmstadt, Germany

- Dissertation in final preparation
- Partial Information Decomposition (PID), information-theoretic representation analysis and agentic workflows for neural and AI systems

2020.04 - Present

Master's by Research in Biosciences: Neuroscience

University College London • London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

2012 - 2014

Bachelor of Science (Honours) in Computer Science

Anglia Ruskin University • Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

2009 - 2012

SELECTED AWARDS & FUNDING

EDTH / 2025

Vision and scene-reconstruction award from the Bundeswehr Cyber Innovation Hub

HUMAN BRAIN PROJECT

Research funding

NEAR HORIZON

Confidential-computing grant

UCL & SHEFFIELD

Competitive academic scholarships

IOS APPLICATIONS

15 international product awards



SIGNATURE

Sepehr Mahmoudian

PLACE Berlin, Germany

DATE 8 August 2026

PUBLIC ENGINEERING PORTFOLIO

GITHUB Open-source projects

github.com/sepahead

HUGGING FACE Models & datasets

huggingface.co/torusprime



SELECTED AI ENGINEERING PROJECTS



engram

RESEARCH PROTOTYPE

EVIDENCE-LINKED MODELS

Turns computational-neuroscience papers into evidence-linked specifications and bounded NEST artifacts, keeping recorded modeling decisions auditable.

Python Knowledge Graphs NEST



manwe

ALPHA

AIRSPACE PERCEPTION

Research workbench for vision, acoustics, multi-camera geometry and tracking; model exports must pass artifact-identity and numerical gates.

Python PyTorch Model Export



pid-rs

V0.9 PRERELEASE

INFORMATION DECOMPOSITION

Decomposes information into shared, unique and synergistic parts, with method-level provenance and explicit research boundaries.

Rust / Python Information Theory Statistical ML



galadriel

RESEARCH PROTOTYPE

CROSS-SENSOR CONSISTENCY

Fail-closed monitor for sensor disagreement on one track, using per-channel innovation tests and signed correlation over producer-declared projections.

Rust NIS / CUSUM Signed Correlation



melkor

HARDENING

GAUSSIAN-SPLAT INTEGRITY

Inspects and deterministically converts 3D Gaussian splats without a GPU, surfacing numeric hazards and field provenance before pipeline use.

C++ PLY / SPZ Asset Validation



prisoma

0.9 CANDIDATE

AUDITABLE VLA EXPERIMENTS

Research toolkit for auditable VLA capture, intervention and replay, with canonical run logs that make claim boundaries and negative results explicit.

Rust / Python VLA Evals Intervention / Replay



crebain

RESEARCH PROTOTYPE

SPATIAL SITUATIONAL AWARENESS

Unifies Gaussian-splat scenes, native object detection and multimodal 3D tracking in one tactical view for simulated drone operations.

Rust Sensor Fusion Native Inference



NCP

V0.8 RELEASED

NEURAL CONTROL PROTOCOL

Project-agnostic protocol linking neural simulators and AI controllers to robots through four QoS planes, with safety-gated action and per-frame provenance.

Rust Canonical JSON Zenoh



haldir

EXPERIMENTAL

MISSION AUTHORIZATION

Checks signed intents against leases, trusted state and policy; only an allow decision can construct a fresh gate-authored command.

Rust Authorization Signed Intents



cobot-atlas

HF downloads 1,775 PUBLISHED + DOI

OPEN ROBOTICS DATASET

Citable robotics atlas of 2,023 unique GLB meshes, from industrial parts to manipulation objects, with its generation pipeline published alongside the data.

Python Hugging Face 3D / VLA

PORTFOLIO & ENGINEERING ACTIVITY



AI / ML ENGINEERING STACK

FIVE CAPABILITY LAYERS

AI SYSTEMS	LLMs / VLMs Generative AI AI Agents Tool Use Model Evaluation Fine-Tuning / SFT
MACHINE LEARNING	Python PyTorch Computer Vision Multimodal AI VLA Evaluation Deep Learning
KNOWLEDGE & RETRIEVAL	Knowledge Graphs Graph RAG RAG Embeddings Vector Search Reranking Data Provenance
SYSTEMS & MLOPS	Rust C++ FastAPI ONNX TensorRT CUDA Metal / Core ML Docker
CLOUD & INFRASTRUCTURE	GCP AWS Azure Kubernetes Terraform GitHub Actions Linux Zenoh